

School of Computer Science and Artificial Intelligence

Lab Assignment # 7.2

Program : B. Tech (CSE)
Specialization : -
Course Title : AI Assisted Coding
Course Code : 23CS002PC304
Semester : II
Academic Session : 2025-2026
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Enrollment No. : 2403A51L45
Batch No. : 52
Date : 30/01/26

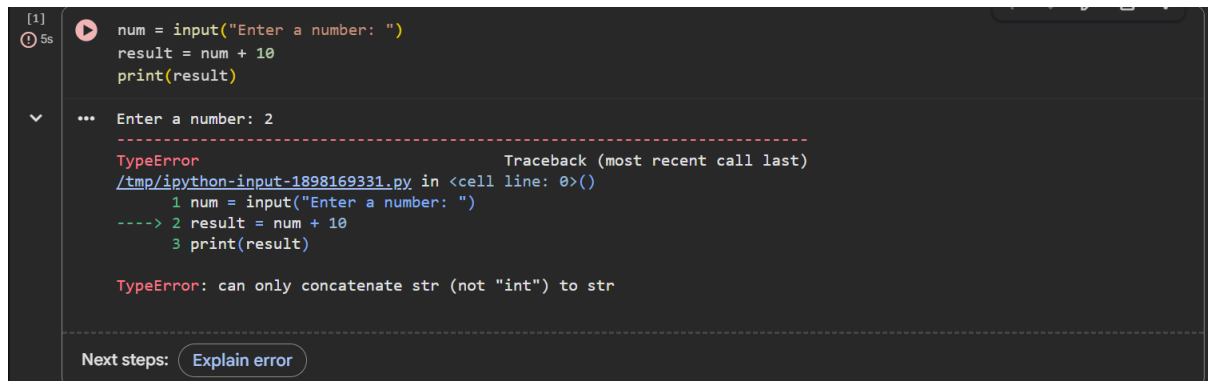
Submission Starts here

Screenshots:

Task 1 – Runtime Error Due to Invalid Input Type

(Buggy Code):

```
num = input("Enter a number: ")  
result = num + 10  
print(result)
```

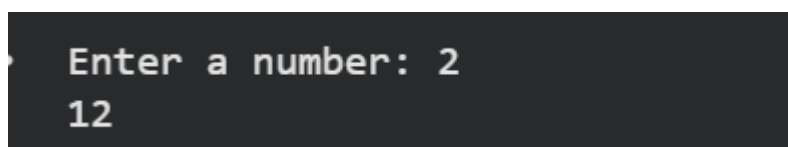


```
[1] 5s num = input("Enter a number: ")  
      result = num + 10  
      print(result)  
  
... Enter a number: 2  
  
-----  
TypeError                                Traceback (most recent call last)  
/tmp/ipython-input-1898169331.py in <cell line: 0>()  
      1 num = input("Enter a number: ")  
----> 2 result = num + 10  
      3 print(result)  
  
TypeError: can only concatenate str (not "int") to str  
  
Next steps: Explain error
```

Output:



```
num = input("Enter a number: ")  
+num = int(input("Enter a number: "))  
result = num + 10  
print(result)
```

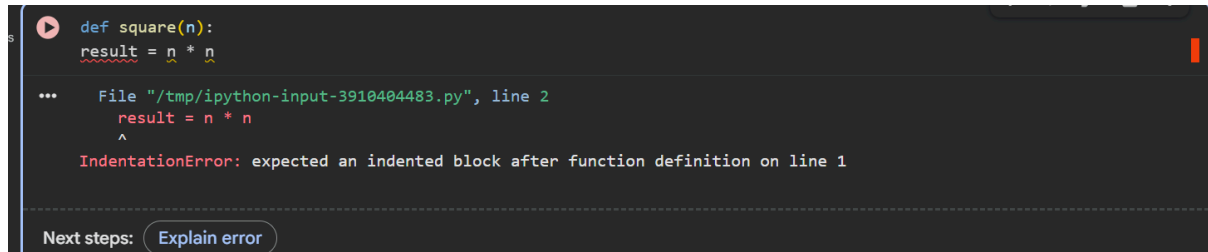


```
Enter a number: 2  
12
```

Task 2 – Incorrect Function Return Value

(Buggy Code):

```
def square(n):
result = n * n
```

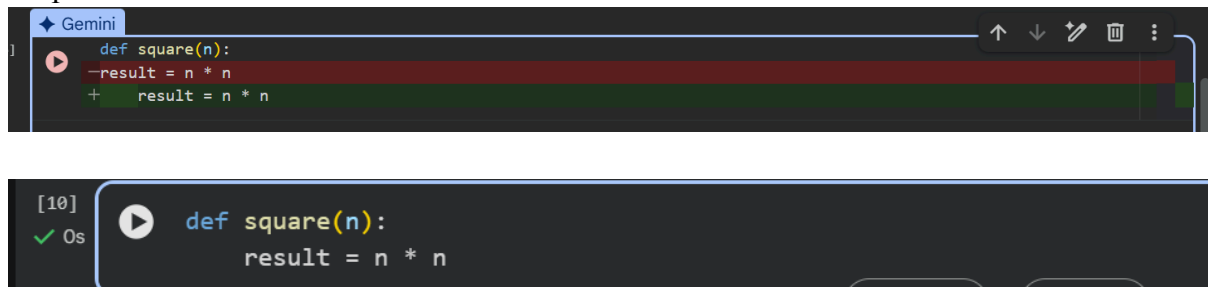


```
def square(n):
    result = n * n
```

File "/tmp/ipython-input-3910404483.py", line 2
 result = n * n
 ^
 IndentationError: expected an indented block after function definition on line 1

Next steps: [Explain error](#)

Output:



```
def square(n):
    result = n * n
```

File "/tmp/ipython-input-726334973.py", line 3
 print(numbers[i])
 ^
 IndentationError: expected an indented block after 'for' statement on line 2

Next steps: [Explain error](#)

Task 3 – IndexError in List Traversal

(Buggy Code):

```
numbers = [10, 20, 30]
for i in range(0, len(numbers)+1):
print(numbers[i])
```

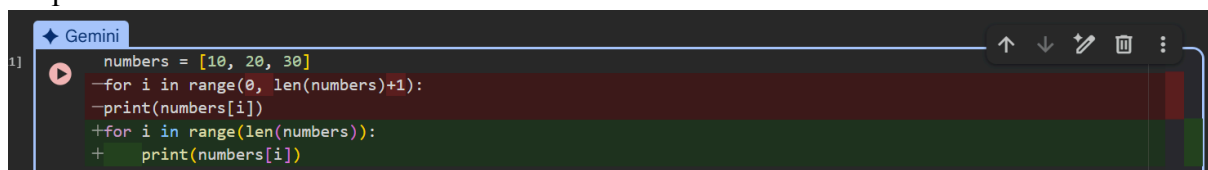


```
numbers = [10, 20, 30]
for i in range(0, len(numbers)+1):
    print(numbers[i])
```

File "/tmp/ipython-input-726334973.py", line 3
 print(numbers[i])
 ^
 IndentationError: expected an indented block after 'for' statement on line 2

Next steps: [Explain error](#)

Output:



```
numbers = [10, 20, 30]
for i in range(0, len(numbers)+1):
    print(numbers[i])
```

File "/tmp/ipython-input-726334973.py", line 3
 print(numbers[i])
 ^
 IndentationError: expected an indented block after 'for' statement on line 2

Next steps: [Explain error](#)

```

... 10
    20
    30

```

Task 4 – Uninitialized Variable Usage

(Buggy Code):

if True:

pass

print(total)

```

[13] 0s if True:
      pass
      print(total)

... File "/tmp/ipython-input-1170978020.py", line 2
      pass
      ^
IndentationError: expected an indented block after 'if' statement on line 1

Next steps: Explain error

```

Output:

```

Gemini
[13] if True:
      -pass
      + pass
      print(total)

```

```

Gemini
if True:
    pass
+total = 0 # Or any other initial value
print(total)

```

```

... 0

```

Task 5 – Logical Error in Student Grading System

(Buggy Code):

marks = 85

if marks >= 90:

grade = "A"

elif marks >= 80:

```
grade = "C"  
else:
```

```
grade = "B"  
print(grade)
```

```
[16]  
0s  
marks = 85  
if marks >= 90:  
    grade = "A"  
elif marks >= 80:  
    grade = "C"  
else:  
    grade = "B"  
print(grade)  
  
... File "/tmp/ipython-input-2691675298.py", line 3  
      grade = "A"  
      ^  
IndentationError: expected an indented block after 'if' statement on line 2  
  
Next steps: Explain error
```

Output:

```
Gemini  
marks = 85  
if marks >= 90:  
- grade = "A"  
+ grade = "A"  
elif marks >= 80:  
- grade = "C"  
+ grade = "C"  
else:  
- grade = "B"  
+ grade = "B"  
print(grade)  
  
... C
```